

The design and evaluation of dialogues for interactive voice response services

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Good dialogue design is an essential component in the delivery of new interactive telephone services. It wins a competitive edge by improving the usability of services and reducing the risks of customer dissatisfaction.

This paper outlines the steps taken to obtain consistent and effective dialogues across the company through the use of the Dialogues Style Guide, dialogue engineering courses and focus groups. Aspects of dialogue design will be illustrated with examples, trials and services developed within BT.

Important results from dialogue studies under BT's Strategic University Research Initiative at the Centre for Communication Interface Research (CCIR) at the University of Edinburgh are reported, together with a description of current activities.

The achievements of the industry-funded Dialogues 2000 initiative is outlined in its pursuit of dialogue consistency. Currently this organisation has a membership of over 217 companies involved with interactive voice response systems.

1. Introduction

Interactive voice response (IVR) systems are becoming all-pervasive within the telephone network. This growth is largely driven by business requirements to differentiate service offerings from competitors. IVR systems allow emphasis to be placed on customer satisfaction by making information and transaction services accessible around the clock, and reducing call times and time spent on hold. Automation contains the costs of service provision, and increases efficiency by freeing up skilled operators to concentrate on higher-value tasks or complex enquiries. As the technology improves, customers are becoming more tolerant of IVR solutions. They are becoming increasingly confident in dealing with machines rather than humans in the performance of functions such as banking and catalogue shopping transactions, or when making enquiries over the telephone.

The effectiveness of an IVR system depends on its functionality, performance, cost, reliability and usability. In recent years, attention has been shifting towards the quality of the system interface that the caller will encounter to guide their interactions. User guides and on-line help may be available, but the dialogue design is often key to a successful IVR system. Experience has shown that there are major benefits to be derived from good human interface design and that this can have a significant impact on the commercial success of automated telecommunications products [1]. Good design improves the usability of voice

services — it reduces overall development costs and the time to roll-out by eliminating problems early in the design cycle. Risks of customer dissatisfaction are also reduced, as well as the likelihood of callers moving to competing products. Streams of revenue can only be sustained if end users are able to achieve their goals and return again and again to use the service.

This paper outlines the approaches BT has taken to design effective dialogues for voice-based services built on research and product trials. It sets out principles that govern good design and exposes problems that commonly occur in dialogues. Finally, areas currently being considered for further research are described.

2. Dialogue design

A good dialogue makes an application easier to use. It makes the users feel comfortable and in control, guides them to give predictable responses, and ensures they know how to get help or exit at any time. It must be clear, simple and consistent to accommodate all users, regardless of their experience or degree of computer literacy.

2.1 Dialogues Style Guide

Within BT, large numbers of people are involved with the conception, design, development and marketing of new

telephone services. The most useful vehicle for the promotion of best practice has been through the mechanism of the Dialogues Style Guide. The guide advocates a design methodology based on accumulated experience, and addresses both DTMF (TouchTone) and speech user interfaces. It addresses requirements capture, dialogue representation, message composition, message recording, prototyping, and dialogue evaluation, and includes a section devoted to screenphones.

The document is World Wide Web based, is interactive and provides audiovisual feedback. Designers are able to experience the advice for themselves. It also supplies links to supporting research reports and dialogue specifications.

2.2 Focus groups

New dialogue problems are constantly being encountered. It is the task of the application engineers in dialogue focus groups to identify such problems and ensure that previously successful solutions are reused if possible. Outstanding problems are prioritised and 'upstreamed' for research as appropriate. At the same time the community is made aware of the latest research findings through reports, presentations and bibliographies. Members of focus groups are requested to review research proposals in their own interest areas to ensure that the work is relevant and is conducted in as realistic a way as possible.

2.3 Dialogue engineering courses

From time to time dialogue engineering courses are run to provide a rapid route for BT people to acquire or improve necessary skills. The courses are based upon the Dialogues Style Guide, and participants are invited to bring their own dialogue problems to act as case studies. The User Centred Design Group also conducts periodic workshops to present methods of conducting service evaluations and analysing the results. These activities emphasise one of the most important tenets of dialogue design — the design review process.

2.4 The quality assurance design review

The dialogue engineer often enters the scene when much of the functionality of the service has been mapped out, either by marketing constraints or by platform constraints. Rarely is the dialogue designer given a clean sheet of paper! The designer serves as a liaison between many people who often have conflicting demands on the product. The review is therefore an essential part of the design process where people from marketing, product management, delivery, customer service and human factors, along with technologists and developers come together and agree on a strategy. Given an agreed dialogue specification, the stakeholders can be more confident that the product being designed is not only close to the customer's

requirements, but also satisfies a multitude of constraints in implementation. A rapid process can be invoked in cases where accumulated experience explicitly dictates the best practice.

2.5 Dialogue evaluation

In spite of painstaking care in the specification of a dialogue there is still no guarantee that it will work in practice! Time spent refining a theoretical dialogue specification suffers from diminishing returns. It is much more important to press on for an early human factors evaluation of the product prototype to understand how people interact with the system.

There are various evaluation techniques, for example:

- an expert evaluation may be conducted by design professionals that are not involved with the development effort — they predict problems based on knowledge and experience of user behaviour,
- focus groups are moderated group discussions that target defined issues — for example, telephone operators would be in the best position to highlight typical customer issues and behaviour when reporting telephone faults,
- usability trials are the most common, and are often used in conjunction with the other techniques — they are designed to identify common problems and paths of navigation through the system, and to measure overall perspectives of effectiveness, efficiency and satisfaction.

An initial usability trial with a few subjects is invaluable in evaluating the initial dialogue design. Four to five participants will expose an estimated 80 percent of the usability deficiencies, allowing major errors to be corrected before further trials are run [2]. The size and scale of the main trial will depend on product functionality and context of usage, as well as the variability in the target population and the evidence already available from similar products.

Subjects participating in a trial are normally selected to be representative of the target market. This may necessitate balancing male/female ratios, for example, and choosing from a range of specified ages. Experience of similar IVR services, geographic background and demographic class may also be considered when assessing potential subjects for specific services.

Trials should be based on a set of scenarios that focus on the principal features that sell the product. Subjects are asked to perform a number of tasks (e.g. 'Phone the service and get a quote for IBM stocks') that a typical user would be likely to want to do and which are designed to evaluate the usability and acceptability of the product. Depending on

time and resource constraints, a one-to-one interview or questionnaire usually follows. There is a trade-off between the depth of information accrued and its ease of analysis. Whenever possible, observing the trialists as they perform the scenarios provides valuable clues to behaviour and attitudes that do not surface during the administration of the questionnaire or interview. In fact, it is common that the trialists struggle throughout the session yet state that the product was easy to use. This may be due to a desire to please, or to cover up feelings of inadequacy. The trials result in recommendations for improvements in the dialogue structure or wording that arise both from an analysis of objective statistics collected by the platform, and subjective feedback from the participants. Early resolution reduces the risk of sending out a product with major flaws, and will reduce future calls to the helpdesk.

3. Some problems and solutions

3.1 TouchTone or speech

A choice must be made whether the dialogue will accept TouchTone or speech input. TouchTone is limited by the penetration of the TouchTone service within the targeted demographic or regional area, the number of options that can be offered at each node, difficulty in negotiating complex cascaded menus, and a restriction on entering data that can be represented numerically. Speech recognition technology is becoming increasingly widespread and offers solutions to these problems, but makes the dialogue design a more complex process. Issues then arise regarding the vocabulary size, performance of the recognisers, and synonym handling.

3.2 Dialogue wording

Some dialogue defects occur under circumstances that are difficult to predict and only manifest themselves during a major trial or after product launch. For example, during the development of the CallMinder [3] service one of the prompts was:

'Who are you calling?'

and expected a name to be uttered by the caller. However, a small percentage of callers were hesitating and saying things like 'By phone!' It was apparent after some investigation that these callers were hearing 'How are you calling?' This highlights the effect of voice accent and style on interpretation. It also emphasises the need to avoid placing key information-bearing words at the beginning of prompts, where they will often not be heard. The use of non-verbal cues should be considered carefully. In one trial approximately 50% of callers remained silent following an opening double beep. Others chatted or made general noise, probably not realising that they could respond, and waited until a second, verbal prompt was played out, thereby

giving the impression that the system was slow. Such spurious utterances were sometimes misrecognised, resulting in the caller ending up in an unexpected part of the dialogue. Such callers often hung up or spent time struggling to extract themselves from the confusion. Interestingly it was observed that callers did not attempt to use DTMF over the non-verbal cue.

Prompt wording greatly affects the responses. Research indicates that callers change their manner of speaking to emulate that of the service. This includes speed of delivery, enunciation, vocabulary, and grammatical construct. For this reason, avoid excessive politeness, which will encourage out-of-vocabulary polite responses. By using clear, closed questions, extraneous speech is discouraged. Also, callers are more likely to use valid keywords if they are included in the surrounding messages.

3.3 Complexity and cognitive load

Complicated dialogue instructions, although accurate and logically correct, are confusing to callers. A trial of a service that invited callers to either speak or use DTMF inputs used the following prompt:

'Please enter the phone number followed by square¹ or say each digit separately after each tone'.

A large percentage of callers hesitated or made mistakes at this point in the dialogue. The problem was alleviated by removing the Boolean logic and separating the different instructions:

'Please enter the phone number followed by square. (pause) Alternatively you may say each digit separately after each tone.'

3.4 Dialogue structure

It has been observed that if the caller experiences problems with the service or becomes confused about what response is expected, then they will tend to remain silent while they work out what to do. It is therefore very important that subsequent prompts help the caller by explaining what is happening and what response is required.

In one trial system a rolling menu dialogue structure was used in which a single available option was listed after each subsequent failure to extract a valid response from the caller (see Fig 1).

A significant percentage of callers became frustrated as they did not know what to say. The dialogue did not help as it just continued giving them the next piece of information, which was not relevant to their immediate needs. This

¹ 'Square' refers to the # symbol.

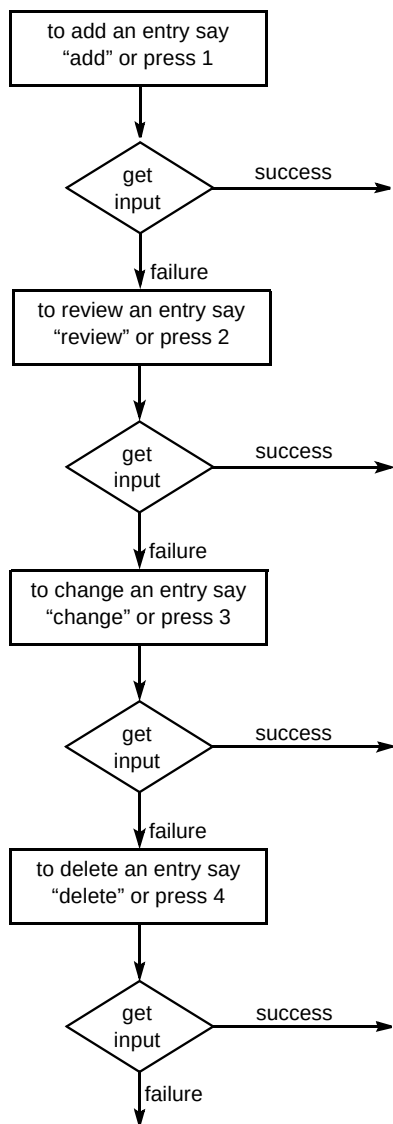


Fig 1 A rolling menu dialogue structure.

tended to result in the caller hanging up, saying random words or shouting — in short ‘an unhappy customer’. Another observation was that some callers did not respond until they heard the option they wanted. The caller seemed unaware that they could respond with any valid option at any of the menu prompts. This resulted in the overall impression that the system was slow.

The dialogue structure was changed to a two-level ‘prompt and answer’ structure where the dialogue initially asked the caller to say a command and then, if they experienced a problem, played out the list of available commands as shown in Fig 2.

The callers were then able to determine which command they required and respond appropriately, significantly improving their experience with the service.

Recent results have revealed that the traditional menu structure used in DTMF dialogues cannot be successfully

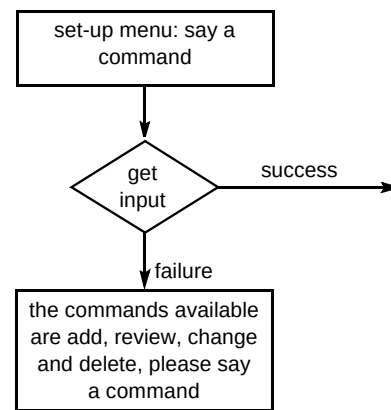


Fig 2 A two-level ‘prompt and answer’ structure.

applied to speech-based dialogues. This does not take advantage of the capability to foster a more natural, conversational interaction that could circumvent much of the menu hierarchy.

3.5 Navigational aids

In complex dialogues it is important to inform the caller which function they are accessing and what commands are available from that point within the dialogue. This helps the caller to navigate through the dialogue successfully, and reduces the occurrence of out-of-vocabulary utterances (i.e. saying the wrong word in the wrong place) in speech recognition services.

There are a number of mechanisms that can be used to aid the caller. In recent trials, implicit confirmation has been used to improve performance and perceived accuracy of the system, thereby improving customer satisfaction. A context-sensitive word or phrase is used. This would often be simply repeating the word recognised, for example:

(System): ‘Main menu. Please say a command.’

(Caller): ‘Specific fund.’

(System): ‘Specific fund. Which fund?’

While this interaction is very easy and quick to include in services, it gives the caller the confidence that the system understood what was spoken. In the case where the system misrecognises an utterance, this has been shown to improve the time in which the caller recovers to achieve their goal.

3.6 Priming

First time callers are inevitably the ‘worst’ performers. This is also the time when they are most likely to form their attitudes to the service and when they are making a crucial decision to return and use the service again or not. It has been shown that the success rate is doubled if callers have first listened to a demonstration of the service.

Good dialogue design goes hand in hand with the design of user guides and scripts for demonstration lines. Further evidence exists that priming material has a significant

influence on caller behaviour. This is supported by recent trials where some of the words recognised by the system were different from words used in priming material sent to subjects. Here the subjects frequently used the words specified in the priming material, even though this meant that they had to ignore what the system was asking for. This also illustrates the importance of consistency across media!

For some services it is not possible to guarantee to provide a caller with information prior to the call. This is especially true for systems that are used infrequently by large numbers of callers. In this case, it is vitally important that the service uses simple, explicit prompting with an approach that is consistent with other services with which callers may be more familiar.

3.7 Habituation

Experiments have shown that callers rapidly habituate to repeated use of a service and this is sustained over extended periods of inactivity. This has implications for the design of certain types of system, especially services that replace existing systems.

It is vitally important therefore, that habituated behaviour is supported within new systems. This does not mean to say that systems cannot be changed. For example, areas of functionality that are infrequently used can be modified, but only with care. However, the major features of services should not be modified without due consideration of the effects on experienced callers.

In a recent trial the keyword 'main menu' enabled the callers to return to the main-menu from most points in the dialogue. It was noticeable that callers soon learnt that it was available and would consistently use it as soon as they hit a problem or got confused. However, this keyword was not globally available and callers were confused when it sometimes appeared not to work.

3.8 The voice

The speaker type and tone of voice used for the system output depends on the type of service. Service impressions are affected by attitudes to voices, for example, based on gender, speed of speech, level of formality, ages, and accents.

Every attempt should be made to record all prompts at one session. It is difficult to reproduce the same intonation from day to day, and it is often necessary to re-record all of the prompts when the dialogue changes. Once callers become accustomed to that voice, future updates should be made with the same speaker if possible.

A choice must be made whether to use pre-recorded speech or text-to-speech (TTS). Recorded speech sounds more natural, but recording each vocabulary item that may need to be spoken may be too time consuming, too difficult to maintain, or the message set may be unknown ahead of

time. Many of BT's speech services combine recorded information with TTS-generated fragments, for example, 'The total of your order is <forty pounds, twelve pence>' with the variable data supplied by the speech synthesiser. BT's synthesis system, Laureate [4], constructs the speech from elemental components of an individual's recorded voice. This preserves the voice characteristics of the original speaker, making the concatenation of recorded and synthesis speech as seamless as possible. It also allows the system to be modified to effect local accents and other languages.

3.9 Foreign translations

Most message prompts for interactive voice response systems can be translated between languages without affecting usability. However, problems occur if the prompt contains a variable such as a time and date stamp. For example, in the BT Chargecard service the customer is prompted in English as follows:

'Delivery status for message <message> recorded at seven oh five on Wednesday the ninth.'

The prompt is constructed in real time by concatenating separate pieces in an appropriate order. However, in Japanese (Romaji) the prompt becomes:

'Tsugi no messeiji <message> no soosin suteitasu rokuo jikan <date> <day> <hours> <minutes>.'

with a different order of day, date and time. This means that the concatenation rules have to be dependent on the language and could add considerable complexity to the dialogue structure if a large number of languages are used. Sometimes such problems can be avoided by simplifying the English, or by instructing the translator to produce translations that fit the required format.

Another example was seen in a German service where the confirmation prompt:

'Call Bob'

was used. However, when used within the German language, this translated to:

'Rufen sie <name> an'.

3.10 Attitude versus recogniser performance

The functional relationship between the level of accuracy of a speech-based service and user attitude towards that service has been characterised in Fig 3.

The graph in Fig 3 indicates that attitudes rise more and more rapidly as 100% recognition is approached. It shows that in the absence of perfect recognition great care must be

taken to restore attitudes through the use of good confirmation and help strategies.

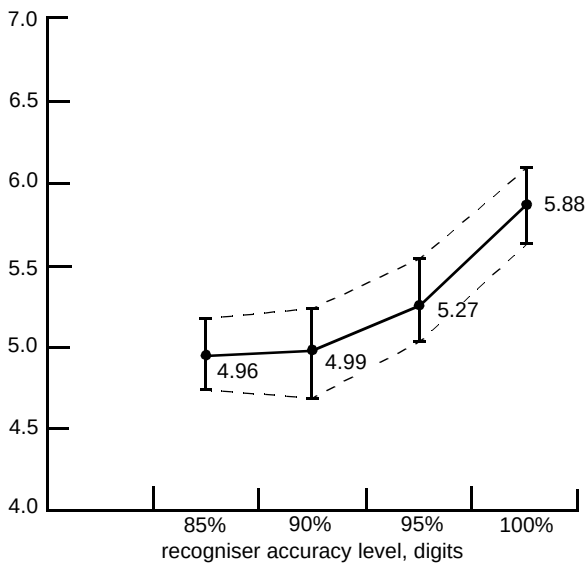


Fig 3 Mean scores on Likert Attitude Scales for each of the four settings of the simulated recogniser with population mean interval estimates at 95% confidence level.

The caller's initial interaction with a system will significantly influence their overall attitude to the recognition performance. It was observed in a trial that callers who experienced difficulty in entering an account number by speech recognition at the start of the dialogue had a very low impression of the overall service, despite achieving near-perfect recognition throughout the rest of their interaction.

In one trial, a poor dialogue that elicited many out-of-vocabulary responses gave the user the impression of low recognition performance. The recogniser was, in actual fact, achieving a high level of accuracy on valid words.

It was also noted that callers who achieved their goals perceived an overestimation of the actual recognition performance. Helping users recover from input or system errors then becomes as important as improving the system accuracy.

3.11 Error-handling

The design must accommodate people that do not respond as instructed. For example, they may speak too soon, pause too long, give incomplete or nonsensical responses, or provide extraneous words. If users detect that an error has occurred, their subsequent strategies are often based on human conversation — those do not always work with IVR systems. They often repeat the request, rephrase it making semantic or syntactic substitutions, add additional information, simplify the request by breaking it into parts, or ignore the prompt and continue to repeat their initial request. Error-handling therefore plays an important role in the formation of caller impressions of the service.

The most effective technique for eliciting an acceptable response has been to provide progressive assistance. Three-tiered messages are recommended, each with increasingly explicit wording and explanation. The first prompt is brief, the second re-worded but still relatively short, and the third contains additional guidance and possibly an example of an expected response.

Callers should always know where they are, what they can do, and how to get help. Context-sensitive help should be available at each interactive stage, and, if possible, callers experiencing problems should be connected with a human operator or given a helpline number.

3.12 Consistency and standards

Customers will perform well and perceive a high standard of service if they encounter standards and protocols they have met before. To this end, a collaborative project, Dialogues 2000 [5], has been set up with the aim of proving and finding agreement on future dialogue standards for IVR services among industrial partners. To date over 215 companies have joined the project. Investigations and reports have been produced on prompting tones, keypad naming and functionality, attitudes to voice types, speaker verification, and spoken digit string grouping.

4. Future research

In current systems, the dialogue flow is mainly system controlled with information being gathered one item at a time through a rigid dialogue structure. Being steered through a dialogue with a machine can be tiresome, and frustrating when the caller gets lost or ends up with the wrong piece of information. New approaches are resulting in the emergence of fluent spoken language systems that allow the caller to respond much more spontaneously, take control of the interaction, and navigate more flexibly through the system. Rather than constantly prompting for explicit information, they use discourse and domain knowledge to resolve ambiguous expressions such as 'give me the next one', or 'does it come in red?' and understand references to earlier information. Since the path through the system cannot be predetermined, and callers may give more than one piece of information at a time, the dialogue becomes difficult to represent. The wordings required are significantly altered and the prompt options are rapidly expanded. In addition, it may be hard for the user to identify the scope and capabilities of the service.

Some work has been performed to test the use of inferences and assumptions to avoid redundant or annoying questions. For example, if a caller requests a reminder call at four o'clock, it is inferred that means that afternoon. If no date is given, the assumption is that the reminder call is for today.

Research into attitudes to dialogues is continuing under BT's Strategic University Initiative at CCIR, Edinburgh University [6]:

- metaphors are being examined as aids to navigation and as a means of improving user performance [7],
- dialogues that make use of a talking head model are being explored for usability,
- dialogue strategies using natural language technology are being researched and their effects studied,
- alternative menu structures are being tested for usability and acceptability,
- investigations into the effect of psychological characteristics of user populations are being performed.

5. Conclusions

The dialogue is the interface with the customer; it is the only means by which the customer perceives an IVR product and plays a significant part in the performance and success of new market entry.

It is essential that application engineers involved with dialogue design are kept aware of best practice, not only to maintain high performance, but also to ensure consistency across products. This encourages habituation which in turn leads to further performance improvements.

It is crucial that the design team understands the needs and requirements of the end user. As a rule dialogues should be kept as simple as possible to reduce the risk of confusion. Defects in dialogues are not always obvious and human factors evaluations are an important weapon in the battle to improve quality.

The move towards global communications will necessarily create a huge demand for dialogues in many languages. Much is understood about English language dialogues but there are sure to be many new challenges for our designers when meeting the requirements of our overseas customers.

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